

18 R&P TEOM 1400, Site Operational Procedures

18.1 Introduction

The R&P TEOM 1400 particle analyser was once widely used throughout the AURN. Since 2008, these have gradually been upgraded to FDMS units and there are no longer any unmodified TEOMs in the network. However, there are unmodified TEOMs at some sites which have been affiliated into the AURN for the other gaseous pollutants monitored there. For this reason, this document has not been removed but remains available. This document describes LSO procedures for the TEOM analyser.

The regular site visits carried out by the LSOs are a major factor in ensuring high quality data from the AURN. The following jobs are done during these site visits:

1. Pre calibration checks;
2. Changing the particulate monitor filter (if applicable);
3. Calibration of analysers;
4. Post calibration checks, safety and security inspection.

As well as these routine jobs, there will sometimes be a need for non-routine site visits – for example in the event of apparent instrument or system malfunction.

Local site operators will be trained in all aspects of normal equipment operation by the CMCU, and in the relevant calibration procedures by the QA/QC Unit. LSOs must keep copies of instrument manuals at each site, and will need to make sure they understand the normal operating principles and characteristics of the instrumentation.

The following sections of the manual describe step-by-step procedures which must be followed during site visits. It is essential that the procedures are followed as written, for both routine and non-routine site visits, to ensure that reliable and accurate air quality measurements are made.

If your site also contains analysers for gaseous pollutants, in addition to the TEOM, you will also need to refer to the operational procedures for the relevant type of analysers.

18.2 Preparation

1. Upon arrival at the site, check the pollutant levels on the front panel of the TEOM to see if an episode is occurring (PM_{10} concentration $> 100 \mu g m^{-3}$). If so, please phone the CMCU. Do not proceed with the calibration as this will result in valuable data from the episode being lost.
2. Take a new set of check and calibration sheets and enter Site, Date and Operator and Start Time.
3. Ensure the Operational Manual is to hand, and follow the instructions carefully.
4. Ensure that the site toolkit (if provided) is complete.

If the instrument has no “out of service” switch, contact the CMCU. This will allow them to ensure that calibration data are not disseminated as ambient data.

18.3 Initial Checks for the TEOM

A number of initial visual checks must be made on the equipment before carrying out any further tasks. If at a site with other analysers for gaseous pollutants, these will form part of the "pre-calibration" checks. Some checks require a tick and some require a value to be recorded on the precalibration sheet. Complete all the checks for all the analysers and ancillary equipment. When all checks are complete, inform the CMCU if any are not correct, before proceeding with the calibration.

The R & P TEOM instrument has a 4 line display screen, as shown in Figure 18-1. The top line is fixed, and displays, from left to right, current status code, current operating mode, percentage of filter lifetime used, current RS-232 mode and current time.

The other 3 lines of the display can be used to scroll through a list of 16 information lines displaying various parameters. Use the cursor keys on the keypad to scroll up ↑ and down ↓. Two lights marked "POWER" and "STATUS" are also visible on the front panel.

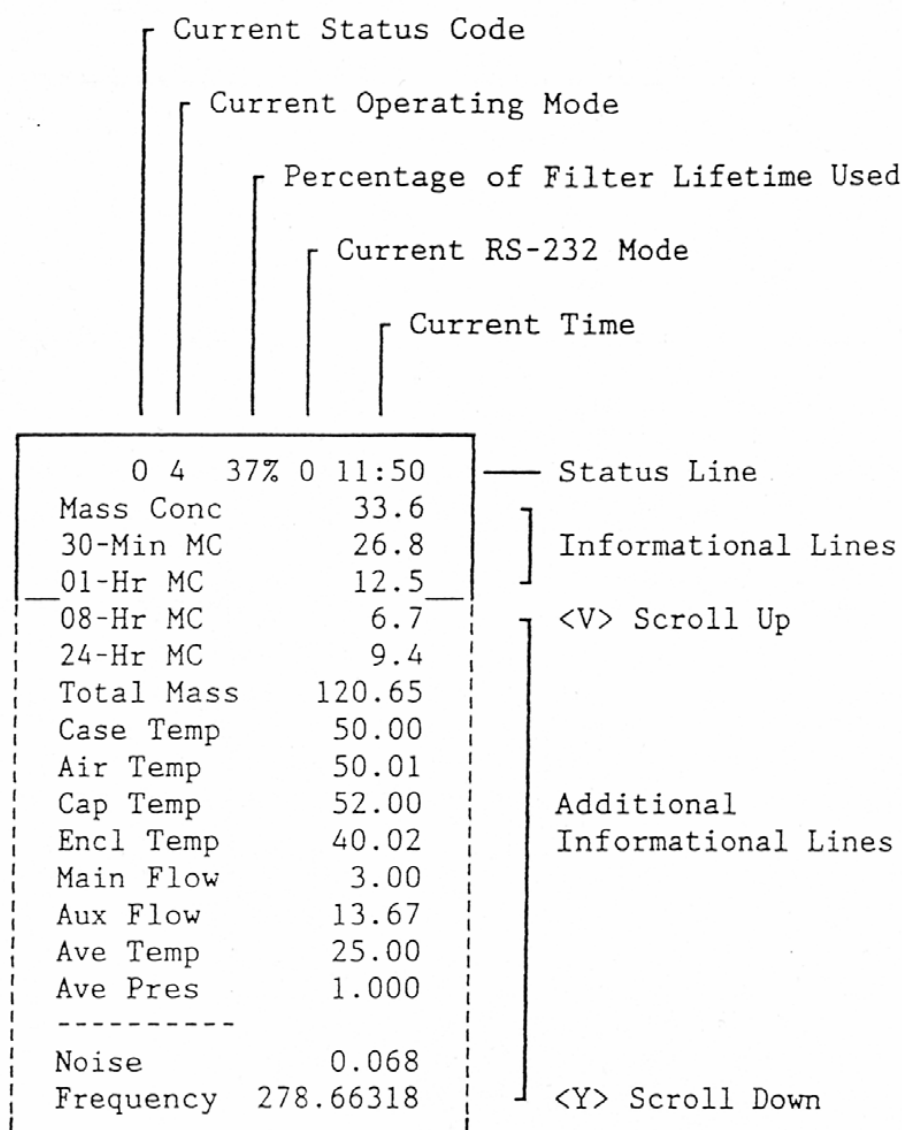


Figure 18-1 TEOM Display Panel

Record the following information on the precalibration checklist:

1. In normal operation, the "POWER" switch will be on and the "STATUS" light off.
2. Check these and record on the pre-calibration checklist.
3. Record current status code, current operating mode, percentage of filter lifetime used, current RS-232 mode and current time from the top line of the display.
4. Telephone the CMCU if the time displayed differs by more than 5 minutes from Greenwich Mean Time.
5. Record the first three information lines in the appropriate space on the precalibration checklist.
6. Press ↓ to scroll through all information lines and record the information.
7. Press ↑ to return to the top of the information lines.
8. Carry out a visual check that the PM₁₀ head is intact.

18.3.1 Modem (where fitted)

Check that the lights on the modem are lit, the 25 way connector is secure, and the phone line is plugged into the BT socket. Record on the pre-calibration checklist. It is likely that the communications for the TEOM are incorporated into that for the other site analysers.

18.3.2 Completion of Precalibration Checks

If any of the above checks are not correct, inform CMCU before proceeding with calibration. If all correct, proceed to next section.

18.4 TEOM Particulate Monitor

The TEOM particulate monitor filter cartridge must be changed every 2 or 4 weeks (as advised by your CMCU) or when the "percentage of filter lifetime used", as shown on the top line of the instrument display is 80% or greater. It is recommended that the TEOM filter cartridge box is stored in the sensor unit of the TEOM analyser so they are pre-conditioned before they are changed with existing filters.

Whenever the filter cartridge is changed, the PM₁₀ head must be cleaned as detailed below. The analyser requires at least one hour to stabilise after filter cartridge changing. Therefore, if the site also contains gaseous pollutant analysers, it is recommended that the PM₁₀ head should be cleaned first, then the gas analysers can be calibrated while you are waiting for the TEOM to stabilise.

18.4.1 Cleaning the PM₁₀ Head

The PM₁₀ head is located on the sample inlet tube above the roof of the monitoring station. To access the PM₁₀ head you will need to use the ladder. Take great care that the ladder is positioned correctly and is secure before stepping onto it. Extra care should be taken if raining as the roof of the station may be slippery when wet. Delay this job if adverse weather conditions (such as ice or snow) mean that it is not safe to use the ladder.

The PM₁₀ inlet needs to be cleaned each time the TEOM filter cartridge is changed, to ensure optimal performance. The cleaning materials required are a small brush, lint free tissues, cotton buds, Decon 90 (1% in H₂O), silicon grease, and distilled water. All components are to be cleaned by soaking Decon 90 on lint-free tissues or cotton buds as appropriate. The component should then be rinsed with distilled water to remove any Decon 90, and dried.

18.4.2 Removing the PM₁₀ Head

1. Switch the TEOM "out of service" switch to ON. If no switch has been explicitly fitted the 'Data Stop' button must be pressed.

2. Carefully lift the complete PM₁₀ head assembly from the TEOM inlet tube.
3. Protect the inlet tube so that rain, snow or dirt cannot enter at any time whilst the head is removed, and take the head inside the monitoring station.
4. Separate the upper and lower inlet halves by unscrewing (counter-clockwise) the acceleration assembly from the collector assembly (see Figure 18-2).

18.4.3 Cleaning the Acceleration Assembly

1. Mark the upper and lower plates of the assembly with a pencil so that the unit can be correctly aligned on reassembly.
2. Unscrew the four Philips screws from the top plate and remove the top plate and four spacers.
3. Clean the top plate, deflector cone, insect screen, internal walls and the underside plate.
4. Inspect the large diameter o-ring for wear and replace if necessary. Wipe any grease off with a tissue, and apply a thin coating of fresh silicon grease to the o-ring and the aluminium threads.
5. Careful reassemble, using the pencil marks to align the top and bottom plates.

18.4.4 Cleaning the Collector Assembly

1. Clean the walls, the three vent tubes and the base of the assembly with a lint-free cloth soaked in Decon 90. Rinse with distilled water.
2. Use cotton buds and Decon 90 to clean the three vent tubes, base of the assembly and weep hole in the collector plate where the moisture runs out to the moisture trap. Rinse with distilled water.
3. Disconnect rain jar assembly from lower collector plate assembly. Clean inside brass tube with cotton buds and Decon 90. Rinse with distilled water.
4. Remove the rain jar and clean. For units with a cork sealing ring inside the cap of the jar, put a thin coating of silicon grease on the gasket and install the jar. If the sealing gasket is neoprene, no silicon grease is required.
5. Reconnect rain jar assembly to lower collector assembly. Ensure rain jar is sitting vertically.
6. Inspect the two inlet tube o-rings for wear and replace if necessary. Wipe off any grease present, and apply a thin coating of fresh silicon grease to the o-rings.
7. Clean the internal threads of the assembly with Decon 90 on a lint-free tissue.

18.4.5 Replacing the Head

1. Screw the Acceleration and Collector assemblies together until the threads are hand tight. DO NOT OVER-TIGHTEN.
2. Place the complete assembly back onto the TEOM inlet tube.

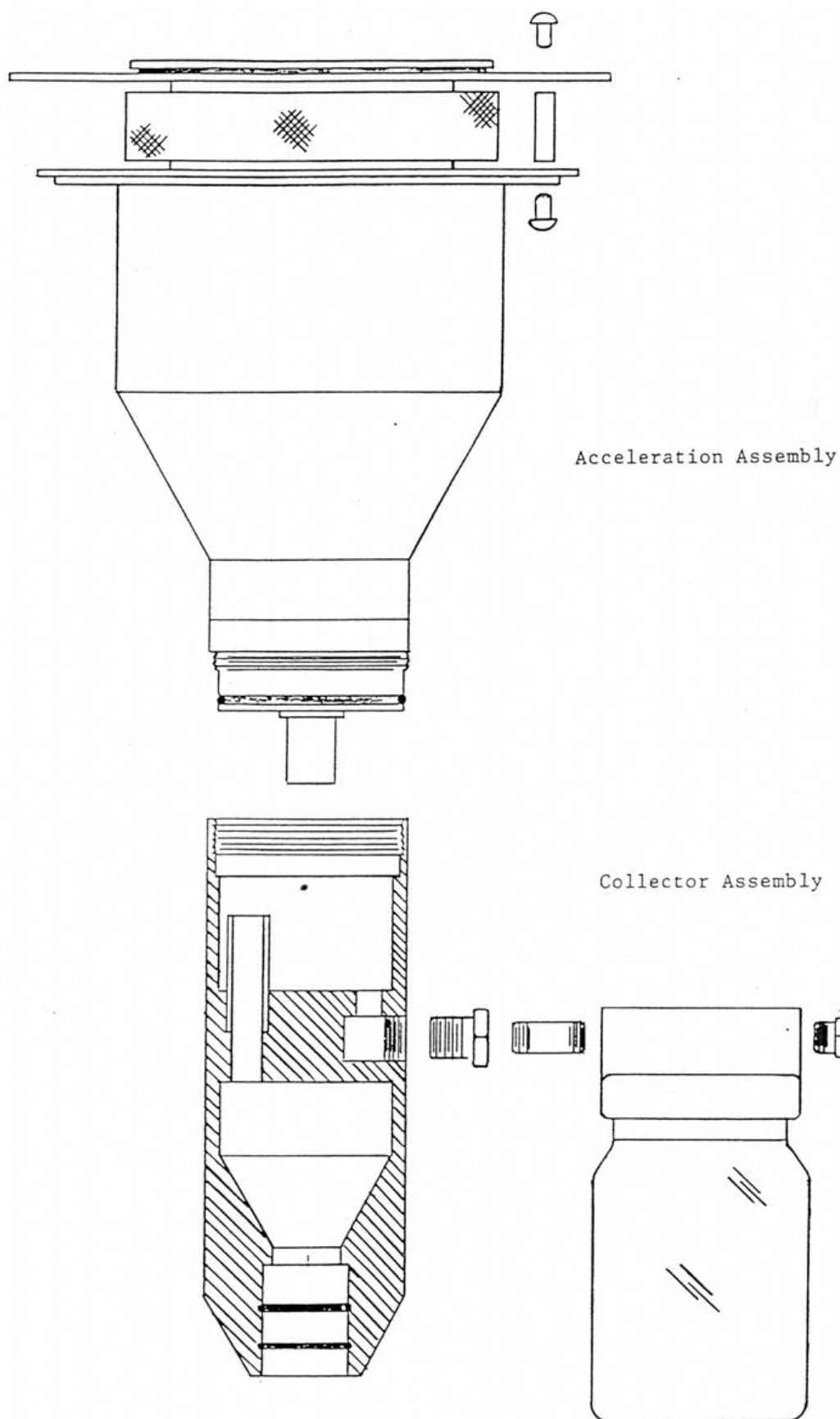


Figure 18-2 TEOM PM₁₀ Sampling Inlet

18.4.6 Filter Cartridge Exchange Procedure

1. Refer to Figure 18-3 when following the instructions for filter exchange. It is recommended that the TEOM filter cartridge box is stored in the sensor unit of the TEOM analyser so they are pre-conditioned before changing. Inform the CMCU when the site stock of unused filters gets down to four.
2. Check that the TEOM 'out of service' switch/'Data Stop' button is ON.
3. Open the door of the TEOM sensor unit.
4. Carefully lift the handle of the mass transducer to swing the transducer into its filter changing position and expose the filter.
5. Carefully insert the filter exchange tool under the filter cartridge so that the filter disk is between the fork and the upper plate of the tool (with the hub of the filter between the tines of the lower form). Gently lift the filter from the tapered element with a straight pull - DO NOT TWIST OR PULL SIDEWAYS.
6. Discard the exposed cartridge and wipe clean the exchange tool with a tissue.
7. Use the exchange tool to remove a new cartridge from the box - DO NOT TOUCH THE FILTER WITH YOUR FINGERS. Note that the box of new filters should be stored inside the TEOM sensor unit, to maintain them at a constant temperature.
8. Hold the new filter in line with the tapered element and lightly insert the hub of the filter onto the tip of the tapered element. Apply a gentle downward pressure to set the filter firmly in place (Figure 18-4) and then carefully retract the exchange tool. Problems with excessive response noise may be experienced if the filter is not seated correctly and firmly on the tip of the tapered element. The filter should, therefore, be positioned with particular care.
9. Gently move the horizontal handle downwards to close the mass transducer; allow the springs to pull it closed for the last centimetre.
10. Close the door of the TEOM sensor unit.
11. After 5 minutes, open the sensor unit and mass transducer again and push down on the filter with the base of the exchange tool. This is to ensure that no movement of the cartridge has occurred during heating of the transducer.
12. Close the door of the TEOM sensor unit.
13. Press <F1> on the TEOM control unit. The TEOM will move through its operational modes as internal parameters stabilise. Check the noise value at the bottom of the display reads <0.1 after 10 minutes. If this is not the case reseal the filter. If this does not reduce the 'noise' sufficiently return to step 1 and repeat the whole procedure. If this is not successful inform the CMCU immediately.
14. Switch the TEOM 'out of service' switch to 'off'. If no out of service switches are fitted the TEOM will return to service automatically on reaching mode 4. If this is not successful inform the CMCU immediately.

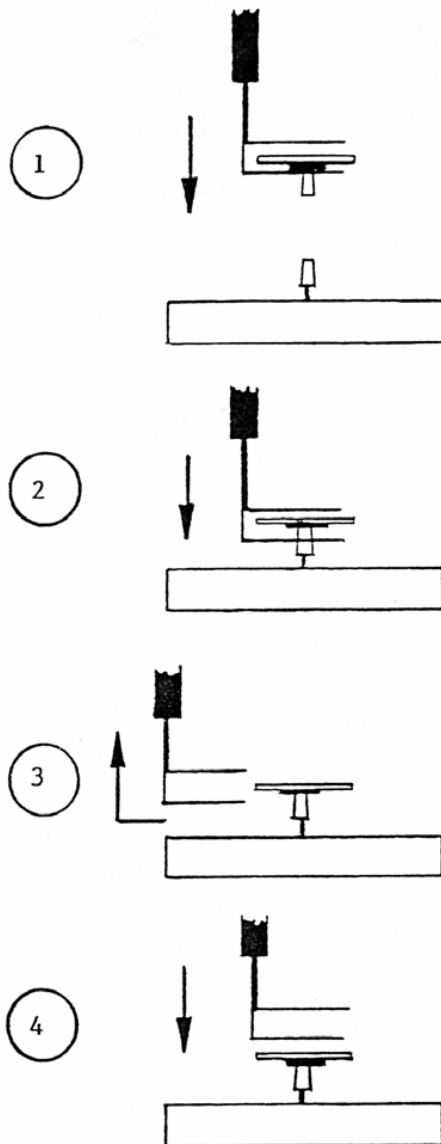
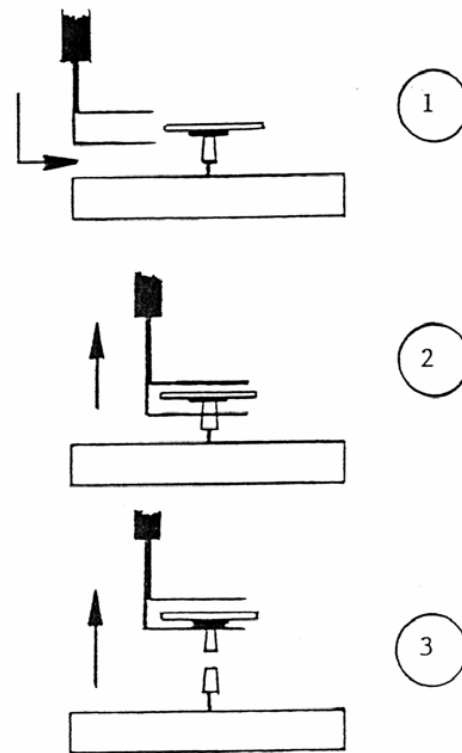
Filter InsertionFilter Removal

Figure 18-3 TEOM Filter insertion and removal

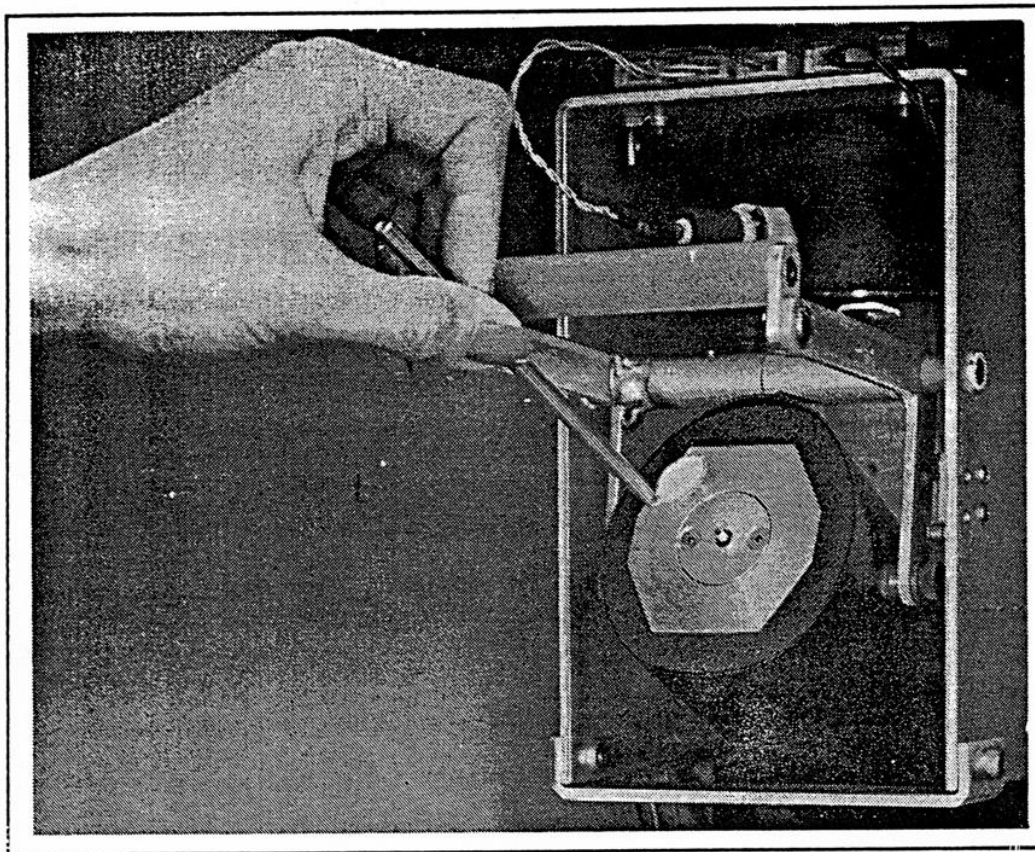


Figure 18-4 TEOM Filter Installation (with Mass Transducer in open position)

18.5 Final Checks, Safety and Security Inspection

As the AURN reports time-averaged concentration data, it is important that operators critically assess the operating condition of the analysers over the time scales used in making discrete measurements. Such assessments may not be possible by consideration of averaged data, as the averaging process may mask such factors as excessive analyser noise or cyclic response changes. Information on analyser performance over very short time periods is important, as this will alert network managers and the QA/QC Unit as to whether instrumentation faults are developing.

Performing calibration checks at regular scheduled intervals, as detailed previously, is an excellent means of assessing instrument performance characteristics. For instance, excessive rise or fall times, possibly due to flow constrictions having developed, will be easily noted by a simple calibration of the analyser. Similarly, "noisy" analyser outputs, which may be caused by inefficient photomultiplier tube cooling systems, will be immediately apparent by observing the analyser output while sampling zero air.

Obviously, the level to which these problems will be detected will depend upon the experience and familiarity with the equipment of each individual operator, but the operator must critically review the calibration he/she has undertaken and comment on any unusual or suspect results or occurrences. In addition, the post-calibration check sheet must be completed as follows:

1. All pre-checks for the TEOM detailed in section 18.3 must now be repeated and recorded on the post-calibration check sheet.
2. Complete the final check section of the post-calibration check sheet. If "out of service" switches are not fitted telephone the CMCU to notify them that the visit is complete so that they may restore data dissemination.

3. Complete the calibration end time.
4. Inspect the cabinet inside and outside for security and safety purposes, paying particular attention to electrical and telephone connections. Check for any signs of vandalism, specially if this may affect safety or lead to a deterioration in data quality. Immediate action must be taken to rectify any situation which may lead to members of the public or monitoring personnel being at risk.
5. Check that the roof area and any fittings are secure. If you have had to access the roof (to clean the inlet of a particulate analyser – see relevant instructions), check that that there are no loose items left on the roof and then stow the ladder safely inside the hut.
6. Where applicable check that all cylinders are firmly closed and the cylinder store locked (Note: if the cylinders are used for autocalibration purposes, via solenoid control, the cylinders must be left on).
7. Ensure the cabinet is clean and tidy.
8. Upon completion of the calibration and on returning to your office, e-mail the Local Site Operator (LSO) Calibration Sheet to the QA/QC Unit at aqadmin@aeat.co.uk and the CMCU at nick.phillips@uk.bureauveritas.com . (These e-mail addresses are provided on the calibration sheet, which will be updated if they change). Keep a copy, so that there is a backup calibration history of all the instruments.